

MAM · Integration and Its Applications — Patterns and Solution Approaches

Based on: marking key + exam reports for 39 past exam questions containing integration from 2016–2025. Original questions are in [topics/02_Integration.pdf](#). Integration is mainly Calculator-free (when it is the main content point, CF 16 questions, CA 7 questions). Also cites 2 questions from the 2016 sample paper ([2016S-F1](#), [2016S-A20](#); original paper in [papers/2016Sample_*](#)). Review status: has undergone one round of independent review and was revised according to feedback (see [Review_log.md](#)).

First, remember: where markers repeatedly deduct marks

- **FTC (Fundamental Theorem of Calculus) is repeatedly called out in reports** (2016, 2017, 2018, 2022, 2025 reports): failing to recognise the various uses of the FTC; the 2022 report specifically calls out the FTC combined with the chain rule (when the upper limit is $g(x)$, multiply by $g'(x)$).
 - **Area $\neq$ integral**: parts below the x-axis must be taken as absolute values, or subtracted in separate intervals (all 4 marks for 2018A-Q16(b) are here).
 - **Write dx after the integral sign** (2023 report: notation mark deduction; 2025F-Q5 awarded 1 mark for "uses correct notation including dx on all integrals").
 - **Areas whose upper/lower boundaries are made up of multiple functions** must be split into multiple integrals (called out in the 2024 report).
 - **Indefinite integrals must include $+ c$** (the key sometimes awards 1 mark separately for this).
 - **"Show that" questions must not skip steps.**
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Pattern 1: Basic indefinite integrals and finding f from f'

Identify: "Determine $\int \dots dx$ "; "If $f'(x) = \dots$, find $f(x)$ given $f(0) = -2$ ".

Approach

1. Memorise the antiderivatives: $\int e^{kx} dx = e^{kx}/k$; $\int \cos kx dx = \sin kx / k$; $\int (ax+b)^n dx = (ax+b)^{n+1}/[a(n+1)]$ ($n \neq -1$); $\int f'/f dx = \ln|f| + c$ (in exams usually $f(x) > 0$, and the key often writes $\ln f(x) + c$).
2. **Evaluating definite integrals** (CF common 3 marks): ✓ antiderivative ✓ substitute upper/lower limits ✓ simplify to an exact value (2017F-Q6(a), 2021F-Q4(b), 2023F-Q1(b), 2023F-Q2(a)(iii)).
3. **When the numerator is a multiple of the derivative of the denominator**: first adjust the coefficient, e.g. $\int (3x+1)/(3x^2+2x+1) dx = \frac{1}{2} \int (6x+2)/(\dots) dx = \frac{1}{2} \ln(\dots) + c$. The key has 1 mark specifically for "modifies the integrand so the numerator is the derivative of the denominator".
4. Find f from f' : integrate and write $+ c \rightarrow$ substitute the known point to find $c \rightarrow$ write the complete $f(x)$. When the answer involves $\ln$, simplify using log laws (e.g. $\ln 32 - \ln 4 = \ln 8$).

Mark-earning points: ✓ correct antiderivative ✓ include constant c ✓ substitute point to find c ✓ write final expression (and simplify as required).

Past exam questions: 2018F-Q3(c)(d), 2019F-Q1(c), 2021F-Q1(c), 2021F-Q4(a), 2022F-Q1(b), 2024F-Q1(b), 2020A-Q10(b).

Pattern 2: "Hence" — using the derivative just found to integrate ★ CF high frequency

Identify: (a) First find $d/dx(x \ln x)$ or $d/dx(2x \sin 3x)$ (2017F-Q8); (b) "Hence show that $\int \ln x dx = x \ln x - x + c$ ".

Fixed three steps (the key usually awards marks for these three steps: 2017F-Q8 and 2022F-Q6 are exactly 3 marks; 2018F-Q7 is split into 4 marks, 2023F-Q2 combines into 2 marks)

1. **Integrate both sides:** $\int d/dx(x \ln x) dx = \int (\ln x + 1) dx$.
2. **Apply the FTC to the left side:** left side = $x \ln x$ (the antiderivative and derivative cancel).
3. **Split the right side and rearrange:** $x \ln x + c_1 = \int \ln x dx + x + c_2 \Rightarrow \int \ln x dx = x \ln x - x + c$ (the two

constants combine into c).

Common errors: writing the conclusion directly; forgetting + c; not splitting the right side using linearity.

Past exam questions: 2016F-Q2(b), 2017F-Q8(b), 2018F-Q7(b), 2022F-Q6(a)(ii), 2023F-Q2(c)(ii). Afterwards there is usually a short part using this result to find an expected value, definite integral or area (2022F-Q6(b)(ii), 2023F-Q2(c)(iii), 2018F-Q7(c)).

Pattern 3: Definite integral = signed area (reading values from a graph)

Identify: A graph made from line segments, semicircles or quarter-circles is given, asking for $\int_0^{12} f(x)dx$, $\int (f(x) - 2)dx$, $\int f'(x)dx$.

Approach

1. Split the region into rectangles, triangles and sectors, with **above the x-axis positive and below negative**.
2. $\int (f(x) - k)dx = \int f dx - k \cdot (\text{length of interval})$.
3. $\int_a^b f'(x)dx = f(b) - f(a)$ (FTC; read the two function values directly from the graph).
4. Find α such that $\int_0^\alpha f = 0$: set the positive area above equal to the negative area below and form an equation.

Mark-earning points: ✓ area split into pieces ✓ add with signs ✓ result (exact value, including π).

Past exam questions: 2016S-F1, 2016F-Q7, 2022F-Q2.

Pattern 4: FTC and functions defined by integrals ★ high frequency (appears in 2016–18, 2021–23, 2025) and repeatedly called out in reports

Identify: $A(x) = \int_a^x f(t)dt$; $d/dx \int_a^x \{g(x)\} f(t)dt = f(x) \cdot g'(x)$; $d/dt \int_t^3 f(x)dx$; "Sketch $A(x)$ "; "Is $F(2\pi)$ positive or negative?".

Core formulas

- $d/dx \int_a^x f(t)dt = f(x)$ (the result must be written as a function of x);
- $d/dx \int_a^x \{g(x)\} f(t)dt = f(g(x)) \cdot g'(x)$ (2022A-Q15: $D(t) = \int_0^{\pi t} \dots \Rightarrow D'(t) = \pi \cdot \sqrt{1+3\cos^2(\pi t)}$);
- When the lower limit is a variable, first swap the limits: $\int_t^3 f = -\int_3^t f$, so the derivative is $-f(t)$ (2022F-Q1(c));
- $\int_a^b f'(x)dx = f(b) - f(a)$ (2018A-Q16(a), 2023F-Q2(a)(iii), 2016F-Q4(c)).
- Linearity + integral function: $\int_2^4 (f + 2)dx = F(4) - F(2) + 4$ (2023F-Q5(b)).

Rules for sketching $A(x)$ (the integral function)

- $A(a) = 0$ (at the lower limit); mark the zeros of A;
- Where f changes from positive to negative, A has a local maximum; where f changes from negative to positive, A has a local minimum;
- Turning points of f correspond to points of inflection of A;
- The increasing/decreasing behaviour of A is determined by the sign of f; mark the value of A at endpoints.

Determining whether $F(x)$ is positive or negative (2025F-Q2): $F(x)$ equals the signed area from 0 to x; compare the positive and negative areas on the graph. The reason must refer to area, not just state the conclusion.

Past exam questions: 2016F-Q5, 2016S-A20, 2017A-Q15(a), 2018A-Q16, 2021F-Q4(c), 2022A-Q15, 2022F-Q1(c), 2023F-Q5(b)(c), 2025F-Q2.

Pattern 5: Area under a curve / between curves

Identify: "Determine the (exact) area bounded by ...", or "Write an integral expression for the area".

Approach

1. Find the intersection points first to use as the limits of integration (the intersection points themselves often earn 1 mark).
2. Determine which is on top: $\text{area} = \int (\text{top} - \text{bottom}) dx$.
3. When finding the area between a curve and the x-axis and the curve crosses the x-axis, or when the upper/lower relationship between two curves swaps, you must split into intervals (when only finding the area between two curves, the curve crossing the x-axis does not by itself require splitting).
4. When the area is inconvenient to integrate with respect to x, use "rectangle – area under the curve", e.g. the area enclosed by $y = e^x$, $y = 2$ and the y-axis = $2 \cdot \ln 2 - \int_0^{\ln 2} e^x dx$ (2017F-Q5, 2019F-Q5).
5. Area involving an inverse function (2020A-Q11(e)): first recognise that the inverse of e^x is $g(x) = \ln x$ (1 mark separately); then note that $\ln 2 \approx 0.69 < 1$, so the region is below the x-axis, hence $\text{area} = -\int_{\ln 2}^1 \ln x dx$ (or write it as $\int_{\ln 2}^1 -\ln x dx$).
6. "Why are the two areas equal": the two functions are translated by the same amount, so the shape is unchanged (2021F-Q5(b) two points: translated in the same direction by the same distance; shape unchanged).

Mark-earning points: ✓ intersection points/limits ✓ correct integral expression (integrand + limits + dx) ✓ antiderivative ✓ exact value.

Past exam questions: 2016A-Q13(d), 2016F-Q6, 2017F-Q5, 2018F-Q7(c), 2019F-Q5, 2020A-Q11(d)(e), 2021F-Q5, 2025F-Q5.

Pattern 6: Definite integral equations with unknown parameters

Identify: "If the area ... is to equal 2, determine the exact value of k"; " $\int_a^b 1/x dx = \ln 3$, find the relationship between a and b".

Approach: Write the integral → use the FTC to obtain an expression involving the parameter → set it equal to the given value → solve the equation (often using log laws: $\ln b - \ln a = \ln 3 \Rightarrow b = 3a$).

Past exam questions: 2017F-Q5(b), 2021F-Q7(a)(ii), 2020A-Q10(d).

Pattern 7: Riemann sums (rectangle estimates)

Identify: Rectangles are drawn on the graph, "demonstrate and explain why $L < \int < U$ ", "best estimate", "state one way to improve".

Approach

1. For the lower sum, use the smaller function value on each subinterval (left or right endpoint, depending on whether the function is increasing or decreasing); for the upper sum, use the larger value; write the expressions separately.
2. **Explanation:** When the function is increasing on the interval, left-endpoint rectangles all lie below the curve, so it is an underestimate; right-endpoint rectangles are an overestimate; when decreasing, it is the opposite (e.g. $1/x$ in 2021F-Q7). When the function is not monotonic on the interval, you cannot simply judge by the endpoints. This explanation is worth 1 mark on its own (the 2017 report noted that explanation-type questions related to area are generally weak).
3. Best estimate = $(L + U)/2$ (trapezium).
4. Ways to improve: increase the number of rectangles (reduce the width); use trapeziums instead; find the function expression and then use integration to calculate the exact value (the key for 2017F-Q9 accepts the first and last items).

Past questions: 2017F-Q9, 2021F-Q7(b), 2024F-Q6.

Pattern 8: Integration applications in context (cross-sectional area, volume, total change)

Identify: building or window cross-sections, ocean trenches, dams, shampoo bottles; or "rate is ... determine the total change".

Approach

1. Volume = cross-sectional area $\times$ thickness/length. Calculate cross-sectional area using integration; units must be consistent (centimetres and metres).
2. Find intersection points or boundaries (e.g. $h(W) = 0$ to find the width, $D(x) = -2$ to find the edge of the ocean trench).
3. Total change = $\int_a^b$ rate of change dt .
4. When the water level h is unknown: write the cross-sectional area as a function of h (the upper limit is an expression in h) $\rightarrow$ cross-sectional area $\times$ length (width) = volume $\rightarrow$ solve for h (2024A-Q16(b): width 4 cm). 2023A-Q14(a) uses the same method to derive the expression for $V(h)$.

Marking points (2023A-Q14(a)): ✓ upper limit expressed in terms of h ✓ correct integral expression ✓ antiderivative ✓ substitute upper and lower limits ✓ simplify to the target expression.

Past questions: 2019A-Q15(c), 2020A-Q10(c)(d), 2022A-Q7, 2023A-Q9, 2023A-Q14, 2024A-Q16. (2021A-Q9 is a similar context, but the whole question does not involve integration.)